

```
SQL> create table pub(pname varchar(30) primary key,address
varchar(30),phone integer);
```

Table created.

```
SQL> insert into pub values('aftab','hubli','1010101010');
```

1 row created.

```
SQL> insert into pub values('husain','dharwad','102010100');
```

1 row created.

```
SQL> insert into pub values('ibrahim','pune','122410400');
```

1 row created.

```
SQL> insert into pub values('umar','mumbai','1111111111');
```

1 row created.

```
SQL> insert into pub values('zameer','hubli','1111122222');
```

1 row created.

```
SQL> select * from pub;
```

PNAME	ADDRESS	PHONE
aftab	hubli	1010101010
husain	dharwad	102010100
ibrahim	pune	122410400
umar	mumbai	1111111111
zameer	hubli	1111122222

```
SQL> create table bok(bid integer primary key,title
varchar(20),pname varchar(30),foreign key (pname) references
pub(pname),pyear integer);
```

Table created.

```
SQL> insert into bok values('1','java','aftab','2022');
```

1 row created.

```
SQL> insert into bok values('2','dbms','aftab','2021');
```

1 row created.

```
SQL> insert into bok values('3','ds','ibrahim','2019');
```

1 row created.

```
SQL> insert into bok values('4','english','husain','2019');
```

1 row created.

```
SQL> insert into bok values('5','html','umar','2021');
```

1 row created.

```
SQL> select * from bok;
```

BID	TITLE	PNAME	PYEAR
1	java	aftab	2022
2	dbms	aftab	2021
3	ds	ibrahim	2019
4	english	husain	2019
5	html	umar	2021

```
SQL> create table baut(bid integer,foreign key(bid) references  
bok(bid),aname varchar(30));
```

Table created.

```
SQL> insert into baut values('1','k murgan');
```

1 row created.

```
SQL> insert into baut values('1','rajesh');
```

1 row created.

```
SQL> insert into baut values('2','pradeep');
```

1 row created.

```
SQL> insert into baut values('3','deppa');
```

1 row created.

```
SQL> insert into baut values('5','wileomson');
```

1 row created.

```
SQL> select * from baut;
```

BID	ANAME
1	k murgan
1	rajesh
2	pradeep
3	deppa
5	wileomson

```
SQL> create table libry(branchid integer primary key,bname  
varchar(30),address varchar(30));
```

Table created.

```
SQL> insert into libry values('11','nehru','hubli');
```

1 row created.

```
SQL> insert into libry values('12','bvb','vidyanager');
```

1 row created.

```
SQL> insert into libry values('13','jss','dharwad');
```

1 row created.

```
SQL> insert into libry values('14','vidyagiri','dharwad');
```

1 row created.

```
SQL> insert into libry values('15','kle','hubli');
```

1 row created.

```
SQL> select * from libry;
```

BRANCHID	BNAME	ADDRESS
11	nehru	hubli
12	bvb	vidyanager
13	jss	dharwad
14	vidyagiri	dharwad
15	kle	hubli

```
SQL> create table bcopy(bid integer,foreign key(bid)references  
bok(bid),branchid integer,foreign key(branchid) references  
libry(branchid),no_of_copies integer);
```

Table created.

```
SQL> insert into bcopy values('1','11','50');
```

1 row created.

```
SQL> insert into bcopy values('1','12','15');
```

1 row created.

```
SQL> insert into bcopy values('2','12','40');
```

1 row created.

```
SQL> insert into bcopy values('5','15','40');
```

1 row created.

```
SQL> insert into bcopy values('4','14','80');
```

1 row created.

```
SQL> select * from bcopy;
```

BID	BRANCHID	NO_OF_COPIES
1	11	50
1	12	15
2	12	40
5	15	40
4	14	80

```
SQL> create table bland(bid integer,foreign key(bid) references
bok(bid),branchid integer,foreign key(branchid) references
libry(branchid),cardno integer primary key,dateout
date,duedate date);
```

Table created.

```
SQL>insert into bland values('1','11','21','01-jan-2023','2-
may-2023');
```

1 row created.

```
SQL> insert into bland values('1','12','22','15-feb-2023','30-
apr-2023');
```

1 row created.

```
SQL> insert into bland values('2','12','23','1-aug-2022','30-
dec-2022');
```

1 row created.

```
SQL> insert into bland values('4','14','24','30-nov-2023','20-
jul-2023');
```

1 row created.

```
SQL> insert into bland values('5','15','25','15-oct-2023','15-
feb-2023');
```

1 row created.

```
SQL> select * from bland;
```

BID	BRANCHID	CARDNO	DATEOUT	DUE DATE
1	11	21	01-JAN-23	02-MAY-23
1	12	22	15-FEB-23	30-APR-23
2	12	23	01-AUG-22	30-DEC-22
4	14	24	30-NOV-23	20-JUL-23
5	15	25	15-OCT-23	15-FEB-23

i. Retrieve details of all books in the library–id, title, name of publisher, authors, number of copies in each branch, etc.

```
SQL> select
b.bid,b.title,b.pname,a.aname,c.no_of_copies,l.branchid from
bok b,baut a,bcopy c,libry l where b.bid=a.bid and
l.branchid=c.branchid;
```

BID	TITLE	PNAME	ANAME	NO_OF_COPIES	BRANCHID
1	java	aftab	k murgan	50	11
2	dbms	aftab	pradeep	15	11
3	ds	ibrahim	deppa	40	11
5	html	umar	wileomson	80	11

ii. Get the particulars of borrowers who have borrowed more than 3 books, but from Jan 2017 to June 2017

```
SQL> select cardno from bland where dateout between '1-
jan-2023' and '30-june-2023'group by cardno having count(*)>=
1;
```

CARDNO
21
22

iii. Delete a book in BOOK table. Update the contents of other tables to reflect this data manipulation operation.

```
SQL> select * from bok;
```

BID	TITLE	PNAME	PYEAR
1	java	aftab	2022
2	dbms	aftab	2021
3	ds	ibrahim	2019
4	english	husain	2019
5	html	umar	2021
6	java	aftab	2022

6 rows selected.

```
SQL> delete from bok where bid='6';
```

1 row deleted.

```
SQL> select * from bok;
```

BID	TITLE	PNAME	PYEAR
1	java	aftab	2022
2	dbms	aftab	2021
3	ds	ibrahim	2019
4	english	husain	2019
5	html	umar	2021

iv. Create a view of all books and its number of copies that a recurrently available in the Library.

```
SQL> create view v_publication as select pyear from bok;
```

View created.

```
SQL> create view v_book as select b.bid,b.title,no_of_copies
from bok b,bcopy c,libry l where b.bid=c.bid and
c.branchid=l.branchid;
```

View created.

```
SQL> select * from v_publication;
```

PYEAR
2022
2021
2019
2019
2021

```
SQL> select * from v_book;
```

BID	TITLE	NO_OF_COPIES
1	java	50
1	java	15
2	dbms	40
5	html	40
4	english	80

Write PL/SQL program to demonstrate implicit cursor.

```
SQL>      declare
2         total_rows number(2);
3         begin
4         update bcopy set no_of_copies=no_of_copies+10;
5         if sql%notfound then
6         dbms_output.put_line('no book selected');
7         elsif sql%found then
8         total_rows:=sql%rowcount;
9         dbms_output.put_line(total_rows || ' book
selected');
10        end if;
11        commit;
12        end;
13        /
5 book selected
```

PL/SQL procedure successfully completed.

```
SQL>      select * from bcopy;
```

BID	BRANCHID	NO_OF_COPIES
1	11	80
1	12	45
2	12	70
5	15	70
4	14	110